

Experiment 2

Analysis of SWR (Standing Wave Ratio) pattern in High Frequency Transmission Lines by varying a) load impedance b) Characteristic impedance

Test case-1

Code-

```
clc;
clear;
close;
Zo = 50;
ZR = 75 - (%i * 25);
K = (ZR - Zo) / (ZR + Zo);
ampK = abs(K);
S = (1 + ampK) / (1 - ampK);
printf("\nReflection Coefficient = %f + %fi", real(K), imag(K));
printf("\nMagnitude of Reflection Coefficient = %f", ampK);
printf("\nStanding Wave Ratio (SWR) = %f", round(S * 1000) / 1000);
```

Output:

Z0=50 ohms variable ZR

Characteristics Impedance	Simulation Result SWR	Theoretical Result SWR
50	1.768	1.77
50	1.808	1.819

[illegible]

CALCULATION OF SWR

Given : $Z_R = 50 - j30 \Omega$

CASE 1: $Z_0 = 60 \Omega$

Reflection coefficient (K) = $\frac{Z_R - Z_0}{Z_R + Z_0} = \frac{(50 - j30) - 60}{(50 - j30) + 60} = \frac{-10 - j30}{110 - j30}$

Multiply numerator and denominator by conjugate (110 + j30)

$$K = \frac{(-10 - j30)(110 + j30)}{(110 - j30)(110 + j30)}$$

$$K = \frac{-200 + j3000}{110^2 + 30^2} = \frac{-200 + j3000}{12100 + 900} = \frac{-200 + j3000}{13000}$$

$$K = -0.0154 + j0.2308$$

Magnitude of K $|K| = \sqrt{(\text{Re}K)^2 + (\text{Im}K)^2}$

$$= \sqrt{(-0.0154)^2 + (0.2308)^2}$$

$$= \sqrt{0.000237 + 0.053269} = \sqrt{0.053506} = 0.2313$$

Standing Wave Ratio (SWR), $S = \frac{1 + |K|}{1 - |K|}$

$$= \frac{1 + 0.2313}{1 - 0.2313} = \frac{1.2313}{0.7687} = 1.602$$

$\therefore$ SWR for $Z_0 = 60 \Omega = 1.602$

CASE 2: $Z_0 = 70 \Omega$

Reflection coefficient (K) = $\frac{Z_R - Z_0}{Z_R + Z_0} = \frac{(50 - j30) - 70}{(50 - j30) + 70} = \frac{-20 - j30}{120 - j30}$

Multiply numerator and denominator by conjugate (120 + j30)

$$K = \frac{(-20 - j30)(120 + j30)}{(120 - j30)(120 + j30)}$$

$$K = \frac{-1500 + j3000}{120^2 + 30^2} = \frac{-1500 + j3000}{14400 + 900} = \frac{-1500 + j3000}{15300}$$

$$K = -0.0980 + j0.1961$$

Magnitude of K $|K| = \sqrt{(\text{Re}K)^2 + (\text{Im}K)^2}$

$$= \sqrt{(-0.0980)^2 + (0.1961)^2}$$

$$= \sqrt{0.009604 + 0.038455} = \sqrt{0.048059} = 0.2192$$

Standing Wave Ratio (SWR), $S = \frac{1 + |K|}{1 - |K|}$

$$= \frac{1 + 0.2192}{1 - 0.2192} = \frac{1.2192}{0.7808} = 1.562$$

$\therefore$ SWR for $Z_0 = 70 \Omega = 1.562$

Test Case 2:

Code-

```
clc;
clear;close;
Zo=200;ZR=650-(%i*475);
K=(ZR-Zo)/(ZR+Zo);
ampK=sqrt((real(K)^2)+(imag(K)^2));
S=(1+ampK)/(1-ampK);
printf("(c)Standing wave ratio = %f",round(S*1000)/1000);
```

Output:

ZR=50/40 ohms variable Z0

Characteristics Impedance	Simulation Result SWR	Theoretical Result SWR
200	5.09	5.03
50	3.500	3.56

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File Help

SciLab 6.1.1 Console

(c) Standing wave ratio = 5.097000
-->

Variable Browser

Name	Value	Type	Visibility	Memory
K	0.641 - 0.2i	Double	local	224 B
S	5.1	Double	local	216 B
Z0	650 - 475i	Double	local	224 B
Zo	200	Double	local	216 B
ampK	0.672	Double	local	216 B

Command History

-- 11/07/2026 10:27:21 -- //

-- 28/07/2026 09:01:54 -- //

-- 06/08/2026 10:18:53 -- //

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